

Capstone Project I: Mutual Fund Analytics

Day 1 Report — Project Setup + Data Ingestion (ETL)

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Git Repository: github.com/anveshtammineni/mutual-fund-analytics

Commit Message: Day 1: Data ingestion complete

1. Project Overview & Repository Setup

This capstone project establishes a complete data analytics pipeline for mutual fund schemes in India. The first day focused on creating a standardized project folder layout, configuring environment dependencies, auditing the initial 10 datasets, validating relations, and retrieving real-time data from the official Open API (mfapi.in).

The directory structure is organized as follows:

- **data/raw/**: Holds the 10 raw CSV datasets and newly fetched API CSVs.
- **src/**: Python execution scripts (data_ingestion.py, live_nav_fetch.py).
- **reports/**: Output summaries and generated PDF documentations.
- **requirements.txt**: Stores the package list and a concise data quality summary.

2. Dependency Installation

A reproducible environment was configured with package pinning. The main dependencies used include:

- **pandas & numpy**: Core data analysis, clean structured loading, and manipulation.
- **matplotlib, seaborn & plotly**: Visualization libraries for subsequent dashboard features.
- **sqlalchemy**: Database connector interface for importing metrics to SQL databases.
- **requests**: Networking client library to query mfapi.in APIs.
- **scipy**: Analytical calculations and statistical metrics.
- **jupyter**: Interactive environments for draft analytics and investigations.

3. Data Ingestion Audit

The script `data_ingestion.py` was executed to read all 10 raw CSV datasets using pandas. Each dataset was audited for shape (rows × columns), column data types, null counts, and duplicates. Below is the audited matrix:

Filename	Description	Shape	Null Values Found
01_fund_master.csv	Fund Master	40 x 15	None
02_nav_history.csv	NAV History	46,000 x 3	None
03_aum_by_fund_house.csv	AUM by Fund House	90 x 5	None
04_monthly_sip_inflows.csv	Monthly SIP Inflows	48 x 6	12 (yoy_growth_pct)
05_category_inflows.csv	Category Inflows	144 x 3	None
06_industry_folio_count.csv	Industry Folio Count	21 x 6	None
07_scheme_performance.csv	Scheme Performance	40 x 19	None
08_investor_transactions.csv	Investor Transactions	32,778 x 13	None
09_portfolio_holdings.csv	Portfolio Holdings	322 x 8	None
10_benchmark_indices.csv	Benchmark Indices	8,050 x 3	None

4. Data Quality & Anomalies Report

During audit, only one dataset presented missing values. In `04_monthly_sip_inflows.csv`, the column `yoy_growth_pct` contains 12 null records. An investigation confirmed that these nulls reside within the first 12 chronological rows of the dataset. Because the calculation of Year-over-Year (YoY) growth requires comparative data from the same month in the prior calendar year, no comparison baseline exists for the first year of entries. This is an expected mathematical limitation, rather than data corruption. No duplicates or unparseable dates were found across the other files.

5. Fund Master Exploration & AMFI Code Validation

A detailed search on `01_fund_master.csv` revealed that the database contains schemes distributed across 10 unique fund houses (e.g. SBI Mutual Fund, HDFC Mutual Fund, ICICI Prudential MF) spanning 2 core categories (Equity and Debt) and 12 distinct sub-categories (e.g. Large Cap, Mid Cap, Liquid, Gilt).

Validation Analysis: Every AMFI code (numeric scheme identifier) present in `01_fund_master.csv` was verified against `02_nav_history.csv`. The script confirmed a **100% referential integrity match**: all 40 unique schemes defined in the Fund Master have corresponding historical NAV listings in the NAV History dataset.

6. Live API NAV Integration

To ingestion live real-time metrics, the script `live_nav_fetch.py` was written to request recent data from the official endpoint `api.mfapi.in`. The results include:

A. HDFC Top 100 Direct Plan (AMFI Code: 125497)

Successfully connected to the GET API and retrieved 3,105 rows. The data was parsed from the JSON response, and dates in the response (originally DD-MM-YYYY format) were converted to SQL-compliant standard YYYY-MM-DD format. The parsed records were written to data/raw/live_hdfc_nav.csv.

B. Key Schemes Verification (5 Major Mutual Funds)

Historical and live NAV records were pulled for 5 major schemes to serve as our project benchmarks. The list includes:

- **SBI Bluechip** (AMFI Code: 119551)
- **ICICI Bluechip** (AMFI Code: 120503)
- **Nippon Large Cap** (AMFI Code: 118632)
- **Axis Bluechip** (AMFI Code: 119092)
- **Kotak Bluechip** (AMFI Code: 120841)

The response was parsed, aggregated, and compiled into a single unified dataset, comprising **16,777 combined records**, saved under data/raw/key_schemes_nav.csv with correct date formatting.

7. Deliverables & Validation Status

All requested Day 1 deliverables have been verified, locally executed, and pushed to your remote repository:

Deliverable	Filename / Location	Status
Directory Structure	data/raw, processed, src, reports, notebooks, sql	COMPLETED
Dependencies list	requirements.txt (with Data Quality summary)	COMPLETED
Ingestion & Exploration	src/data_ingestion.py	COMPLETED
Live API NAV Loader	src/live_nav_fetch.py	COMPLETED
Data Quality Summary	reports/data_quality_summary.md	COMPLETED
Ingestion & API Data	data/raw/live_hdfc_nav.csv & key_schemes_nav.csv	COMPLETED
Version Control Push	GitHub Repository (Day 1 Commit)	COMPLETED

End of Day 1 Report